SAFETY DATA SHEET

NANOtij ® Black Ink

1. Identification

Product identifier NANOtij® Black Ink

Other means of identification Black Pigmented Aqueous Ink

Recommended use Thermal Ink Jet Printing Ink

Company identification Digital Ink Technologies Pty Ltd

Factory 1&2, 74 Mason Street
Campbellfield VIC 3061
Australia
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2. Hazards Identification

GHS Classification of the product

Acute toxicity, Oral (Category 4)
Serious eye damage/eye irritation (Category 2A)

Label elements

GHS07 GHS008

Signal word Warning

Hazard statements

H302 Harmful if swallowed.
H319 Causes serious eye irritation.
H360 May damage fertility or the unborn child.
H373 May cause damage to organs (Kidney) through prolonged or repeated exposure if swallowed.

Precautionary statement (Prevention)

P264 Wash hands and skin thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P280 Wear protective gloves/eye protection/face protection.

Precautionary statement (Response)

P301 + P312 + P330: IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell. Rinse mouth.
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P308 + P313 IF exposed or concerned: Get medical advice/attention.
P337 + P313 If eye irritation persists: Get medical advice/attention.

Precautionary statement (Disposal)

P501 Dispose of contents/container in accordance with applicable local, regional, national, and/or international regulations.

Other hazards None

3. Composition/ Information on Ingredients

Hazardous Components (Chemical Name)	CAS Number	%	EC No./ EC Index No.	GHS Classification
Ethylene glycol	107-21-1	<5	203-473-3 603-027-00-1	Acute Tox. O. 4: H302 STOT RE 2: H373
Diethylene glycol	111-46-6	<10	203-872-2 603-140-00-6	Acute Tox. O. 4: H302
2-Pyrrolidinone	616-45-5	<5	210-483-1	Eye Irrit. 2A: H319 Repr. 1B: H360
1,2-Hexanediol	6920-22-5	<=2	230-029-6	Eye Irrit. 2A: H319
Additives	Proprietary	<0.5	N/A	

4. First-Aid Measures

Inhalation
Move person to fresh air immediately.
If symptoms persist, get immediate medical attention.

Skin contact
In case of contact, take off immediately all contaminated clothing and wash skin thoroughly with soap and plenty of water. Wash clothing separately before reuse.
Get medical advice/attention.

Eye contact
In case of eye contact, remove contact lens and rinse immediately with plenty of water for at least 15 minutes, lift lower and upper eyelids occasionally.
Get medical advice/attention.

Ingestion
Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Rinse mouth thoroughly with plenty of water. If the material is swallowed, get medical attention or advice immediately.

Most important symptoms/ effects, acute and delayed
See section 2 and section 8.

5. Fire-Fighting Measures

FLASH POINT:
>200°F (>93.3°C) (closed-cup).

Suitable extinguishing media
Suitable extinguishing media: water spray, dry chemical or carbon dioxide (CO_2).

Unsuitable extinguishing media
For this mixture, no limitations of extinguishing agents are given.

Specific hazards arising from the chemical
Carbon oxides

Specific protective equipment and precautions for firefighters	Firefighters should wear self-contained breathing apparatus and protective clothing to prevent contact with skin and eyes. Avoid run off into storm sewers and ditches which lead to waterways.
Fire-fighting equipment/instructions	Move containers from fire area.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures	Restrict access to area. Keep unprotected personnel upwind of spill. Exercise appropriate precautions to minimize direct contact with skin or eyes and prevent inhalation of vapors or mists.
Environmental precautions	Contain contaminated water/firefighting water. Transfer to suitable disposal container. Do not release into soil/sewers/drains/surface waters/groundwater.
Methods and materials for containment and cleaning up	Not available.

7. Handling and Storage

Precautions for safe handling	Avoid contact with skin, eye and clothing.
Conditions for safe storage, including any Incompatibilities	Keep out of reach of children. Keep away from excessive heat or cold.

8. Exposure Controls/Personal Protection

Control parameters

Components with workplace control parameters

Exposure controls

Appropriate engineering controls Use in a well-ventilated area.

Personal protective equipment

Eye/face protection	Chemical safety goggles
Hand protection	Protective gloves
Skin protection	Not available
Respiratory protection	Not available

General hygiene Considerations	Handle in accordance with good industrial hygiene and safety practice. Wash thoroughly after handling.

9. Physical and Chemical Properties

Appearance	Black liquid
Odor	Slight odor
Odor threshold	Not available
pH	7.0-10.0
Melting point/freezing point	Not available
Initial boiling point and boiling range	>=100°C
Flash point	Not available
Evaporation rate	Not available
Flammability (solid, gas)	Not applicable
Solubility	Soluble in water
Partition coefficient: n-octanol/water	Not available
Auto-ignition temperature	Not available
Decomposition temperature	Not available
Viscosity (cPs)	Not available

10. Stability and Reactivity

Reactivity	No hazardous reactions with proper storage and handling
Chemical stability	Stable in normal conditions
Possibility of hazardous reactions	Not known
Conditions to avoid	Heat, freeze
Incompatible materials	Strong oxidizing agents
Hazardous decomposition	Carbon dioxide (CO_2), Carbon monoxide (CO)

products

11. Toxicological Information

Symptoms related to the physical, chemical and toxicological characteristics Not available

Information on toxicological effects

Acute toxicity/Effects

Components	Type of value	Species	Test results
Ethylene glycol (CAS 107-21-1)			
Inhalation	LC50	Rat	>2.5 mg/L, 6 Hours
Oral	LD50	Rat	4700 mg/kg
Oral	LD50	Mouse	5500 mg/kg
Dermal	LD50	Rabbit	10626 mg/kg
Diethylene glycol (CAS 111-46-6)			
Oral	LD50	Rat	12565 mg/kg
Dermal	LD50	Rabbit	11890 mg/kg
2-Pyrrolidinone (CAS 616-45-5)			
Oral	LD50	Rat	6500 mg/kg
1,2-Hexanediol (CAS 6920-22-5)			
Oral	LD50	Rat, male	5339-6470 mg/kg
Oral	LD50	Rat, female	6166 mg/kg

Likely routes of exposure

Ingestion Harmful if swallowed.

Skin irritation/corrosion Contact may cause skin irritation.

Inhalation May cause respiratory irritation.

Serious eye damage/ eye irritation	May cause serious eye irritation.
Sensitizing effects	Based on available data, the classification criteria are not met. Not classified.
Specific target organ toxicity -single exposure	Based on available data, the classification criteria are not met.
Specific target organ toxicity -repeated exposure	Based on available data, the classification criteria are not met.
Aspiration hazard	Based on available data, the classification criteria are not met.
Carcinogenicity	Based on available data, the classification criteria are not met.

12. Ecological Information

Toxicity

Ethylene glycol (CAS 107-21-1)

	Type of value	Species	Test results
Toxicity to fish	LC50	Oncorhynchus mykiss (rainbow trout)	18500 mg/l, 96 Hours
	LC50	Leuciscus idus (Golden orfe)	>10000 mg/l, 48 Hours
	NOEC	Pimephales promelas (fathead minnow)	39140 mg/l, 96 Hours
Toxicity to daphnia and other aquatic invertebrates	LC50	Daphnia magna (water flea)	41000 mg/l, 48 Hours
	NOEC	Daphnia (water flea)	24000 mg/l, 48 Hours
	EC50	Daphnia magna (water flea)	74000 mg/l, 24 Hours

Diethylene glycol (CAS 111-46-6)

	Type of value	Species	Test results
Toxicity to fish	LC50	Pimephales promelas (fathead minnow)	75200 mg/l, 96 Hours

2-Pyrrolidinone (CAS 616-45-5)

	Type of value	Species	Test results
Toxicity to fish	LC50	Danio renio (zebra fish)	4600 - 10000 mg/l, 96 Hours
Toxicity to daphnia and other aquatic invertebrates	EC50	Daphnia magna (water flea)	>500 mg/l, 48 Hours
Toxicity to algae	ErC50	Desmodesmus subspicatus (green algae)	>500 mg/l, 48 Hours

1,2-Hexanediol (CAS 6920-22-5)

	Type of value	Species	Test results
Toxicity to fish	LC50	Pimephales promelas (fathead minnow)	72860 mg/l, 96 Hours
Toxicity to daphnia and other aquatic invertebrates	LC50	Daphnia magna (water flea)	>100 mg/l, 48 Hours

Other information Not available

13. Disposal Considerations

Disposal instructions Do not flush into surface water or sanitary sewer system.
Do not dispose of together with general office waste.
Treat or dispose of waste in accordance with all local, state/provincial, and national regulations.

14. Transport Information

DOT Not regulated as dangerous goods.

IATA Not regulated as dangerous goods.

IMDG Not regulated as dangerous goods.

ADR Not regulated as dangerous goods.

Further Information Not a dangerous good under DOT, IATA, ADR, IMDG, or RID.

Environmental hazards No

15. Regulatory Information

US federal regulations US EPA TSCA Inventory: All chemical substances in this product comply with all rules or orders under TSCA.

TSCA Section 12(b) Export Notification (40 CFR 707, Subpt. D) Not regulated.

US OSHA Specifically Regulated Substances (29 CFR 1910.1001-1050) Not on regulatory list.

CERCLA Hazardous Substance List (40 CFR 302.4) Not on regulatory list.

NIOSH-Ca (National Institute for Occupational Safety and Health) Not on regulatory list.

US state regulations

US Massachusetts RTK-Substance List	Ethylene glycol (CAS 107-21-1) 2-Pyrrolidinone (CAS 616-45-5)
US New Jersey Worker and Community Right-to-know Act	Ethylene glycol (CAS 107-21-1) 1,2-Hexanediol (6920-22-5)
US Pennsylvania RTK-Hazardous Substance	Ethylene glycol (CAS 107-21-1) 2-Pyrrolidinone (CAS 616-45-5) 1,2-Hexanediol (6920-22-5)
US Rhode Island RTK	Ethylene glycol (CAS 107-21-1)
US California Proposition 65-Carcinogens & Reproductive Toxicity (CRT): Listed substance	Not on regulatory list.

16. Other Information

Issue date	17-Oct-2024
Revision date	-
Version #	02
Disclaimer	The information contained herein is, to the best of our knowledge and belief, accurate. Since the condition of handling and use is beyond our control, we assume no liability for loss or injury resulting from the use of this product or the information herein. All chemicals may present unknown health hazards and should be used with caution. Safe use of this product is the responsibility of the user.